

ARI 410/510 Project Phase 1: Data Collection & Annotation Setup

University of Michigan-Flint

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(See course Canvas page for due date)

1 Project Phase Overview

Now that you have decided on a task and planned how you will collect your data, it is time to begin with data collection and prepare for data annotation. You will complete the following steps during this phase of the project:

1. **Collect unlabeled data:** Collect data that fits within the constraints of your task. Remember: in general, your data should *not* already be labeled, but it should be possible for a (average or trained) person to determine the label by looking at (or reading/listening to/watching) the data.
2. **Develop annotation guidelines:** Write the set of rules that another person could use to label your data, and follow these rules to label a portion of your data yourself. You should set up a simple interface to annotate the data, designed so that *multiple* annotators can label the same items (see Section 3.3).
3. **Present your task and annotation setup** to the class. This is your first project presentation and helps confirm your setup makes sense before annotation begins (Section 5).

Generated-labels teams (approved by permission): your deliverables for this phase differ — see Section 3.4.

2 Data Collection

You should now begin collecting your data. *Do not save this step for the day before the due date! It may take time to ensure you can get the data that you need.* This will look different for every project, but you should aim to store your data in an easy-to-access way. A simple recommendation is to have a folder where each file is one data point, or, depending on the type of data, you can

have the entire dataset in a `.csv`, `.tsv`, or `.json` file where each row represents a data point.

While collecting your dataset, pay attention to any terms-of-service from the platforms/sources from which you collect data. If you are using an existing dataset, check the license (if one exists). For your own dataset, you should also select an appropriate license, which is sometimes dictated by the license of your source data. You can check out the Creative Commons License Chooser to help with this.

Finally, make sure to document the dataset well in a README file (written in markdown or plain text). This should include details about the source of your data and specifically how it was collected in a way that is detailed enough that someone else could read and follow the same procedure and end up with the same dataset. This file should also clearly explain the format of the dataset (what each file represents, what the columns in a csv file represent, etc). Describe what each instance in the dataset represents (e.g., one movie review, one photo taken from a traffic camera, etc.), how many total instances there are, the day(s) when the data was collected, how data was sampled from a larger set, and any potentially missing data. Note that this README will help prepare you to create a complete Datasheet for your dataset later in the project. **Finally, and importantly, please include an estimate of how long you anticipate it would take for someone to label each item in the dataset** (e.g., can it be done in 5 seconds based on a quick glance, or does it require careful reading and thinking, perhaps taking a minute or longer per instance?). A realistic per-item time tells you how much of your data can actually get multiply annotated. Since you need to annotate some of the data within your team, you can time yourself while doing that as a way to help with this estimation.

3 Annotation Guidelines and Interface

Now that you have your data, you should prepare for annotations. *These will be completed by other students in the class* unless you have identified other individuals who are willing to help with this task. You need to create both the *guidelines* and the *interface* that annotators will use to label the data. At least some of your data needs to be annotated by someone outside of your team who is seeing your guidelines for the first time when they begin annotation.

3.1 Guidelines

Your annotation guidelines should be a written document which explains the following to a potential annotator:

1. A concise overview of the annotator's job
2. The set of labels that can be applied with short descriptions of their meaning

3. Unambiguous (to the extent possible) rules or criteria that should be followed in order to label the data.
 - **Note:** While you should try to be unambiguous, do not try to use these rules to “solve” the ML version of the task or trivialize it to the point that human annotations wouldn’t be needed. E.g., “if the value in the ‘length’ field is greater than 10, label the item as ‘long’ ” *is not a rule that requires a human annotator*. Applying that rule would be a waste of your classmates’ time. On the other hand, something like “label the image with ‘car’ if the image contains any kind motor vehicle that would typically have four wheels, including minivans, SUVs, and Pickup Trucks. If the object is occluded, still apply the label if you are confident, and provide your confidence score using the provided slider” aims to be specific, but still requires human judgment.
4. At least 1 example of each label along with an explanation for why that is the correct label (and if helpful, why the other labels would *not* apply to that instance).
5. Contact information where annotators may connect with you if they face issues

Note that this should be written in a way that it can be understood by the annotators regardless of their background knowledge of your dataset or project. It should not require you to have a meeting with the annotators to explain the task. To give you an idea, some examples of annotation guidelines can be found in section A.7 of this paper, section A of this paper, sections A.2 and A.3 of this paper, and this website.

A note on expert-leaning tasks: if your labels require domain knowledge, invest extra effort in scaffolding — worked examples, decision rules, and hints — so that a classmate with no background can handle the common cases. If your task genuinely cannot be labeled by a non-expert even with good guidelines, talk to the instructor: you may need to recruit your own annotators (allowed), or your task may be a better fit for another data-collection path.

3.2 Interface

You must set up a **real annotation interface** for your task and write complete instructions for how someone could use it to label your data. **A bare spreadsheet is not sufficient** — use a purpose-built annotation tool. A real interface presents one item at a time, records labels cleanly, and produces a structured output file, which avoids the copy-paste errors and inconsistency that spreadsheets invite once several annotators are involved.

Good free, open-source options (pick one that fits your data type):

- **Potato** — recommended default. Zero-code YAML configuration, launches a local web server, covers text and several multimodal task types, and has annotator quality-control features built in. (Developed at UMich.)
- **Label Studio** (Community Edition) — flexible for mixed modalities (text, image, audio, video). Install with `pip install label-studio` and run `label-studio start`. Steeper learning curve than Potato; compute agreement yourself in Phase 2 of the project (the free edition doesn't include it).
- **doccano** — simple and clean for text-only tasks (classification, NER, seq2seq).
- **labelme** or **CVAT** — for image annotation.

These run as local web servers, so you do **not** need to host anything live: you can hand annotators your config plus a short set of setup instructions (install, drop in the config/data, run one command) and they run it locally for their hour. If you'd rather annotators install nothing, hosting one is welcome but optional. Either way, **test your setup end-to-end yourself first**, and make sure your instructions state what output file the annotator produces and how to send it back to you.

If you are into agentic programming (e.g., with Claude Code, Codex) or really *want* to taken on an extra “manual” coding task, you could produce your own custom annotation html page that reads your data, presents it along with the options for labels, and stores users' selections in a standard format. Please reach out to the instructor if you are interested and we can even provide you with some Claude Code credits to use. If you choose to do this, please *confirm your tool works before the deadline*, and in your deliverables, include some documentation of how you prompted the model and a description of what you created and how it works.

3.3 Multiple annotations per item and dividing up the data

Your setup must produce more than one annotation for at least some items. This is not optional: in Phase 2 you will compute inter-annotator agreement and derive ground-truth labels, and both require items that were labeled by multiple people. If every item is seen by only one annotator, you will have nothing to compute.

Who annotates your data. Each student in the class annotates for 2 other teams, so with our enrollment **each team can expect about 5 external annotators**, each spending roughly 1 hour. In addition, **at least one member of your own team must serve as an internal annotator** and label a portion of your data (you should start this early, even before you release your guidelines, and it doubles as a check that your guidelines actually work). Later, you will compare your team's own annotations against the external annotators'.

Plan around a realistic labeling rate. Estimate how many instances one person can label in an hour (this is the per-item time recorded in your README), and use that to size everything below — roughly 5 external hours of labeling are coming your way, plus whatever your internal annotator contributes (at least 1 hour).

The recommended design has two parts:

- **A shared “agreement set.”** Choose a subset of items that *every* assigned annotator (external and internal) will label. This overlapping set is what you use to measure agreement, and having several labels per item lets you take a majority vote for ground truth. You decide how much redundancy you want here.
- **A distributed remainder.** Any remaining annotation effort is spread across the rest of your data for coverage, divided among annotators so you maximize how much of the dataset gets labeled.

Worked example. Say you estimate an annotator can label 60 items in an hour, and you expect 5 external annotators plus 1 internal. You designate a 30-item agreement set that all 6 label (6 labels per item → strong agreement estimate + majority vote). Each of the 5 external annotators then labels 30 further, non-overlapping items for coverage (150 more), and your internal annotator adds coverage too. You finish with 180+ labeled items, 30 of them multiply annotated. Need a larger agreement set? Have everyone label 60 shared items and skip the coverage split. **The tradeoff between redundancy (better agreement / ground truth) and coverage (more labeled data) is yours to make** — just justify your choice.

3.4 If you are on the generated-labels path (by permission)

If your proposal was approved for programmatic label generation instead of human annotation, this phase looks different. In place of annotation guidelines and a peer annotation interface, you should:

1. **Document your label-generation method** in enough detail that someone else could reproduce it (the analog of annotation guidelines): what tool/measurement produces the label, exact settings, and how raw output maps to your label set. If you used code, include a compressed archive of the code or a link to a GitHub repository.
2. **Set up and run a hand-validation** of a random sample of your data to estimate the accuracy of the generated labels. Do not look at the labels, first, rather, look at the data points and write what the correct label should be (so you avoid biasing yourself). Alternatively, randomly flip 50% of the labels before verifying, and ensure you are able to identify the instances where the label was flipped (without looking at which ones were actually flipped). However you set this up, your team should contribute

at least a total of one person-hour hand-validating. This sample check is your stand-in for inter-annotator agreement in Phase 2, so record it carefully.

You still produce the dataset, README, and license as above, make a presentation in class, and your team members still annotate other teams' data in the next phase.

4 Changes from Your Proposal

Your proposal was a plan, not a contract. Based on the feedback you received and on what you learned while actually collecting data and building your annotation setup, you are *encouraged* to refine your task. A better-scoped or more interesting version is better than staying locked into a suboptimal proposal. Along with this phase, submit a short **“Changes from proposal”** note (can be just a few sentences or a bullet list) describing what you changed and why. If nothing changed, say so in a sentence and still include this note. Making appropriate changes is seen as a positive rather than a sign that you did anything “wrong” in your proposal.

Note: you may even change aspects of your task itself, such as the set of labels, after looking at some of your data. This is fine as long as you lock this in before you do your internal annotation. It is possible that while trying to do the internal annotation, you realize adjustments that need to be made. You may do this, but you should always have data annotating using the final annotation scheme that you will ask your classmates to follow.

5 First Project Presentation: Task + Annotation Setup

At the end of this phase you will give a short (5 minutes maximum per team) presentation to the class covering:

1. **Your task**, why it matters, and an example data point.
2. **Basic descriptive statistics about the dataset you collected** — e.g., how many instances, the data type/format, where it came from, and a simple distribution or two that you *can* compute before annotation (text length, source breakdown, time span, etc.), plus your estimated per-item labeling time.
3. **A live look at your annotation interface and guidelines** (generated-labels teams: your generation + validation setup instead).

This serves two purposes: it lets the class see the tasks they will soon annotate, and it provides a chance to confirm every setup makes sense **before** annotation begins, so problems get caught early.

6 Deliverables

To complete this phase of the project, you must do the following by the deadline:

1. Submit, on Canvas, a PDF document *or* link to GitHub repository that contains:
 - (a) A link to your dataset (google drive, dropbox, etc.) which has permissions set such that UM students can access it.
 - (b) a description of the dataset (if using GitHub, this can be in the README.md file; see Section 2 for more on what to include here)
 - (c) the license you plan to use for the dataset
 - (d) your annotation instructions (or link to a web page containing them) — or, for generated-labels teams, your documented generation method + validation plan
 - (e) a short “Changes from proposal” note (Section 4)
 - (f) (conditional) *if* you created a custom annotation platform, e.g., using Claude Code, a note documenting the process you used, how it works, and prompts used.
2. Complete this Google Form to make your annotation request.
3. Create a post in #project-annotation-tasks on Discord with your presentation slides (PDF attachment or accessible link to online version, see Section 5 for more about the presentation) and give your presentation in class on the due date.

Only one submission needs to be made *per team* for all three components.